Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21312

F-412

B.C.A. (Third Semester)

Examination, 2023-24

(New Course)

Paper - BCA-302

Computer Organization

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt **all** sections as directed.

Section-A

Note : Attempt any **five** questions. All questions carry equal marks.

1. (a) Describe various number systems with suitable example.

   (b) How addition and subtraction of numbers is performed using signed 2's complement procedure? Explain with proper example.

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7. (a) Describe the concept of programmed initiated I/O mode of data transfer.

   (b) Draw the work diagram of DMA and illustrate its overall working.

8. Write in brief about the following: (any four)

   (i) Construction of ALU

   (ii) Types of instruction

   (iii) EPROMs and EEPROMs

   (iv) Virtual Memory

   (v) Boolean algebra

   (vi) Cycle Skaling concept

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2. (a) Explain the floating points arithmetic in detail.

(b) Multiply (4) and (–5) using booth's algorithm where (–5) is the multiplier and (4) is multiplicand. Show each step very clearly following the algorithm.

3. (a) What are various types of processor organization? Explain each.

(b) Show the evaluation of following arithmetic expression using stack (step-by-step) : 2*3 + 4*5

4. (a) What do you mean by "Addressing Modes"? Describe them in brief.

(b) Differentiate SRAM and DRAM clearly.

5. (a) Draw and explain the memory hierarchy.

(b) Explain the working of "cache-Memory" with well labelled diagram.

6. (a) How paging is different from segmentation?

(b) What is bus arbitration? Explain its types.

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